Date: 26/09/ 25

Lab Practical #14:

Implementation of parity bit check Using C/Java language with example.

Practical Assignment #14:

1. **C/Java Program: Implementation of parity bit check Using C/Java language.**

```java
import java.util.Scanner;

public class ParityCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // input the binary data
        System.out.print("Enter binary data (without parity bit): ");
        String data = sc.next();

        // choose even or odd parity
        System.out.print("Enter parity type (even/odd): ");
        String type = sc.next();

        int count = 0;
        for (char c : data.toCharArray()) {
            if (c == '1') count++;
        }

        // calculate parity bit
        int parityBit;
        if (type.equalsIgnoreCase("even")) {
            parityBit = (count % 2 == 0) ? 0 : 1;
        } else {
            parityBit = (count % 2 == 0) ? 1 : 0;
        }

        System.out.println("Data with parity bit: " + data + parityBit);
```

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```java
// now checking received data
        System.out.print("Enter received data (with parity bit): ");
        String recv = sc.next();

        int ones = 0;
        for (char c : recv.toCharArray()) {
            if (c == '1') ones++;
        }

        boolean valid;
        if (type.equalsIgnoreCase("even")) {
            valid = (ones % 2 == 0);
        } else {
            valid = (ones % 2 != 0);
        }

        if (valid)
            System.out.println("No Error in transmission");
        else
            System.out.println("Error detected in transmission");

        sc.close();
    }
}
```